Ch 9.1: Maximum Margin Classifier

Lecture 26 - CMSE 381

Prof. Elizabeth Munch. Substitute: Rachel Roca

Michigan State University
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Dept of Computational Mathematics, Science & Engineering

Mon, Nov 11, 2024

Announcements

Last time:

- Ch 8: Random Forests

This lecture:

- Maximal Margin Classifier
- No jupyter notebook for this class

Check the example, there are some typos and it tripped me up this time. Edit 2024 - Didn't find it so I think this was from 2 years ago? **Announcements:**

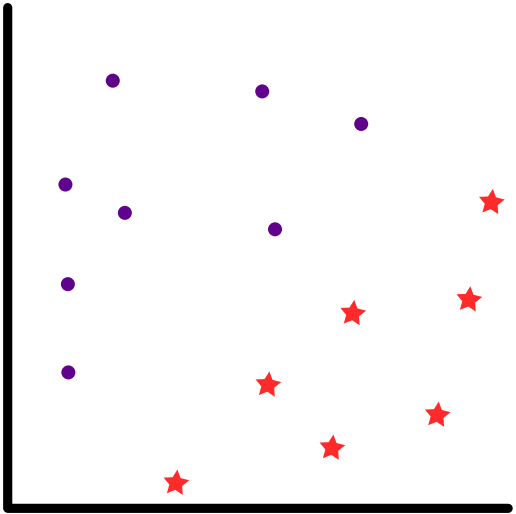
- HW #8 Due Sunday 11/17

Lec #	Date			Reading	HW
21	Mon	10/28	Polynomial & Step Functions	7.1,7.2	
22	Wed	10/30	Step Functions; Basis functions; Start Splines	7.2 - 7.4	
23	Fri	11/1	Regression Splines	7.4	HW #6 Due Sun 11/3
24	Mon	11/4	Decision Trees	8.1	
25	Wed	11/6	Class Cancelled (Dr Munch out of town)		
26	Fri	11/8	Random Forests	8.2.1, 8.2.2	HW #7 Due Sun 11/10
27	Mon	11/11	Maximal Margin Classifier	9.1	
28	Wed	11/13	SVC	9.2	
29	Fri	11/15	SVM	9.3, 9.4	HW #8 Due Sun 11/17
30	Mon	11/18	Single layer NN	10.1	
31	Wed	11/20	Multi Layer NN	10.2	
32	Fri	11/22	CNN	10.3	HW #11 Due Sun 11/24
33	Mon	11/25	TBD: Unsupervised learning/clustering	12.1, 12.4?	
	Wed	11/27	Virtual: Project office hours		
	Fri	11/29	No class - Thanksgiving		
	Mon	12/2	Review		
	Wed	12/4	**Midterm #3**		
	Fri	12/6	No class - EGR Design Day		Project due

Section 1

Maximal Margin Classifier

The goal



- Point out today is all about classification
- Goal is to find a line in the middle that separates the two classes. Lots of options, but we want the one that stays as far from everyone as possible.
- For today, need to use $y \in \{-1, 1\}$ for this to make sense.

What is a hyperplane?

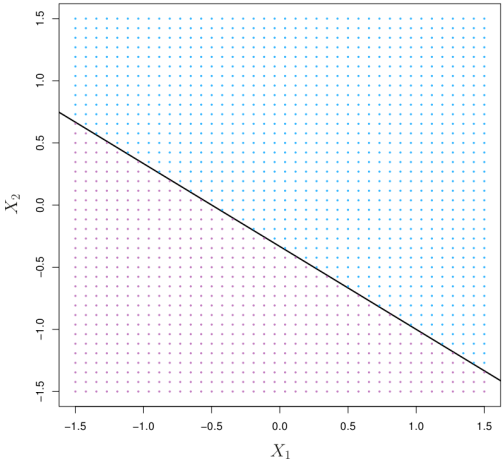
- In p-dimensional space, a hyperplane is a flat affine subspace of dimension $p - 1$
- Draw example for $p = 2, 3$
-

Mathematical definition of a hyperplane

$$\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p = 0$$

- Being on the hyper plane means having a point $X = (X_1, X_2, \cdots, X_p)^T$ satisfying this
- Show that when $p = 2$, this is just a line

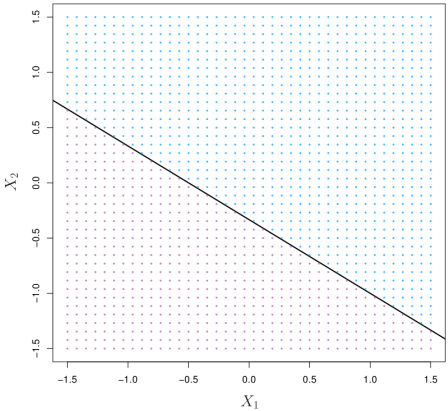
Hyperplane for $p = 2$



$$1 + 2X_1 + 3X_2 = 0$$

There are two sides to every hyperplane

$$\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p < 0 \qquad \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p > 0$$



Classification Setup

Data matrix:

$$X = \begin{pmatrix} - & x_1^T & - \\ - & x_2^T & - \\ & \vdots & \\ - & x_n^T & - \end{pmatrix}_{n \times p}$$

Observations in one of two classes, $y_i \in \{-1, 1\}$

$$Y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$$

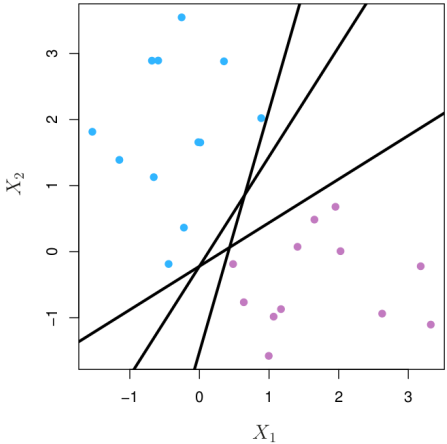
$$x_1 = \begin{pmatrix} x_{11} \\ \vdots \\ x_{1p} \end{pmatrix}, \cdots, x_n = \begin{pmatrix} x_{n1} \\ \vdots \\ x_{np} \end{pmatrix}$$

Separate out a test observation

$$x^* = (x_1^* \ \cdots \ x_p^*)^T$$

Separating Hyperplane

$$\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} > 0 \text{ if } y_i = 1$$
$$\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} < 0 \text{ if } y_i = -1$$



- Strict requirement
- No points can be on the wrong side of the line

Another way to say it

$$\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} > 0 \text{ if } y_i = 1$$
$$\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} < 0 \text{ if } y_i = -1$$

For all i:

$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}) > 0$$

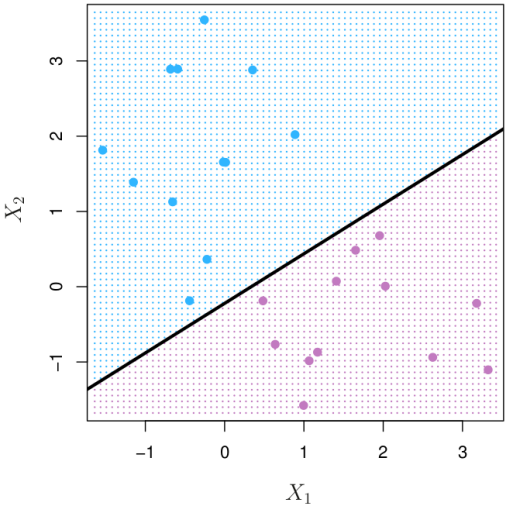
Separating hyperplane becomes a classifier

If you have a separating hyperplane:

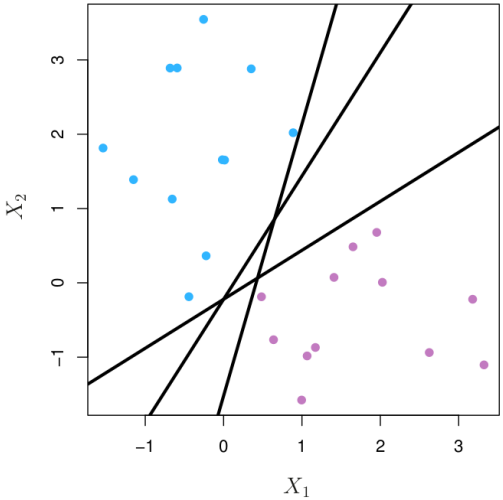
- Check

$$f(x^*) = \beta_0 + \beta_1 x_1^* + \beta_2 x_2^* + \cdots + \beta_p x_p^*$$

- If positive, assign $\hat{y} = 1$
- If negative, assign $\hat{y} = -1$

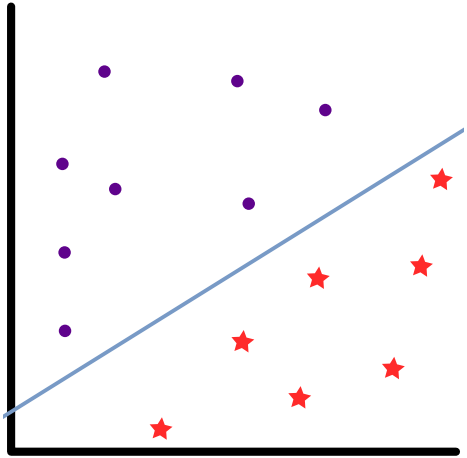


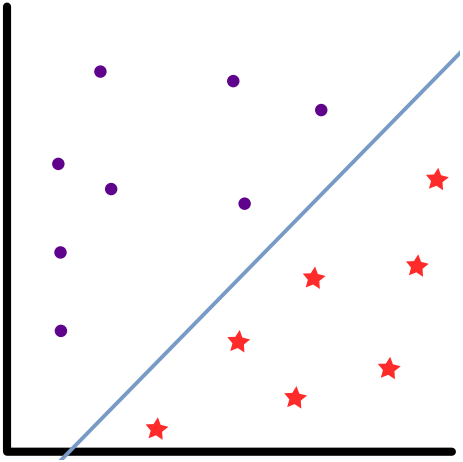
How do we pick?



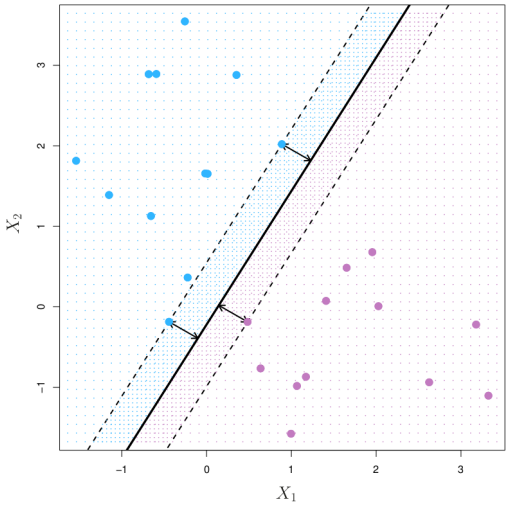
- *There's lots of separating hyperplanes here, so how do we find a "best" one?*
- Compute the perpendicular distance from each training observation to a given separating hyperplane
- Smallest such distance is the *margin*

Distance from an observation to a hyperplane



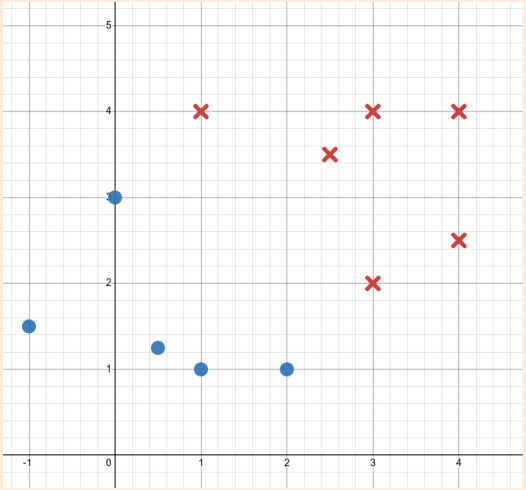


Maximal margin classifier



- *Maximal margin hyperplane* is the hyperplane with the largest minimum distance to the training observations
- Classify using it gives you the *maximal margin classifier*
- Observations that are closest are called *support vectors*
- If points moved slightly then MMH would move
- No effect if any other points moved

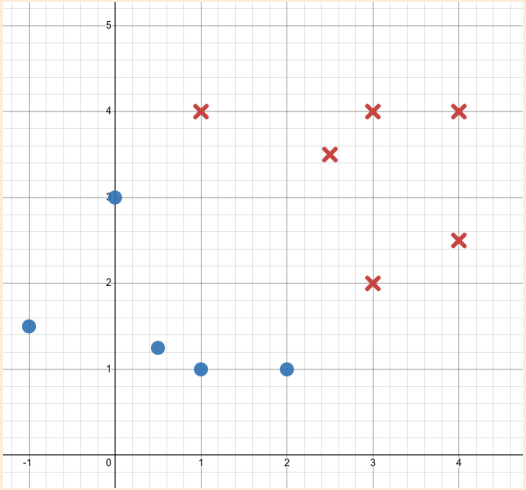
Example



- Sketch the maximal margin hyperplane.
- What is the equation of this line in the form $\beta_0 + \beta_1 X_1 + \beta_2 X_2 = 0$?
- Circle the support vectors. What is their distance from the line?

desmos.com/calculator/lqms253gfq

Extra work space



- $X_1 + X_2 + 4 = 0$
- Points are at distance $\frac{\sqrt{2}}{2}$. Build the isoceles right triangle with d on two sides and 1 for hypotenuse.

desmos.com/calculator/lqms253gfq

Mathematical Formulation

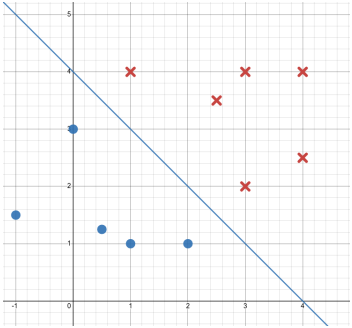
$$\underset{\beta_0,\beta_1,\ldots,\beta_p,M}{\text{maximize}} \; M$$

$$\text{subject to } \sum_{j=1}^{p} \beta_j^2 = 1,$$

$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}) \geq M \;\; \forall\, i = 1, \ldots, n$$

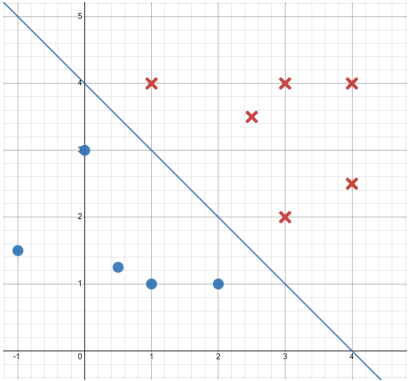
- Last eq forces points to be on the correct side of the hyperplane if $M > 0$
- Making M as big as possible is the maximal margin hyperplane

First constraint



- Had the equation $4 + X_1 + X_2 = 0$
- Can divde by any number and get the same line.
- $\sum_{i=1}^{p} \beta_i^2 = 1^2 + 1^2 = 2$
- We get the same sloped line if we divide by $\sqrt{(\sum \beta_i)}$,
- Just need to solve for the intercept
- $-\frac{4}{\sqrt{2}} + \frac{1}{\sqrt{2}} X_1 + \frac{1}{\sqrt{2}} X_1 = 0$
- $-2\sqrt{2} + \frac{\sqrt{2}}{2} X_1 + \frac{\sqrt{2}}{2} X_1 = 0$
- Check that $\sum_{i=1}^{p} \beta_i^2$ is now 1
- Note that $y = 1$ is Red X, $y = -1$ is blue dots below the line

Second constraint

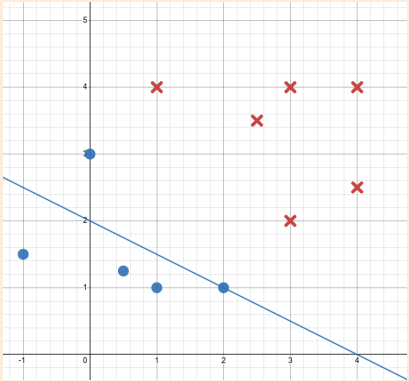


- Blue circles: $y_i = -1$
- Red Xs: $y_i = 1$
- $-2\sqrt{2} + \frac{\sqrt{2}}{2}X_1 + \frac{\sqrt{2}}{2}X_1 = 0$

$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2}) \geq M$$

- For point $(3, 2)$:
  - $y_i = 1$
  - $1(-2\sqrt{2} + \frac{\sqrt{2}}{2} \cdot 3 + \frac{\sqrt{2}}{2} \cdot 2) = \frac{\sqrt{2}}{2}$
- For point $(0, 3)$:
  - $y_i = -1$
  - $-1(-2\sqrt{2} + \frac{\sqrt{2}}{2} \cdot 0 + \frac{\sqrt{2}}{2} \cdot 3) = \frac{\sqrt{2}}{2}$
- For point $(1, 1)$
  - $y_i = -1$
  - $-1(-2\sqrt{2} + \frac{\sqrt{2}}{2} \cdot 1 + \frac{\sqrt{2}}{2} \cdot 1) = \sqrt{2}$

An example with a bad choice of hyperplane

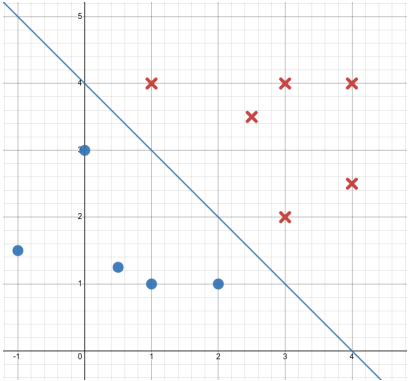


- Blue circles: $y_i = -1$
- Red Xs: $y_i = 1$
- $-\frac{4}{\sqrt{5}} + \frac{1}{\sqrt{5}}X_1 + \frac{2}{\sqrt{5}}X_1 = 0$

What is $y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2})$ for the point $x_i = (0, 3)$?

- $-1(-\frac{4}{\sqrt{5}} + \frac{1}{\sqrt{5}} \cdot 0 + \frac{2}{\sqrt{5}} \cdot 3)$
- $-1(-4 + 6)\frac{1}{\sqrt{5}}$
- $\frac{-2}{\sqrt{5}}$
- Note this is negative, so no positive M will make it so this is a valid line

Second constraint extra space



- Blue circles: $y_i = -1$
- Red Xs: $y_i = 1$
- $-2\sqrt{2} + \frac{\sqrt{2}}{2}X_1 + \frac{\sqrt{2}}{2}X_1 = 0$

$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2}) \geq M$$

- Ensures each on the correct side for a positive M
- When we have unit normal coefficients, the second constraint is distance from the input point to the diagonal.
- Making sure every distance is bigger than M and then make M as big as possible

Mathematical Formulation

$$\underset{\beta_0,\beta_1,\ldots,\beta_p,M}{\text{maximize}} \; M$$

$$\text{subject to} \; \sum_{j=1}^{p} \beta_j^2 = 1,$$

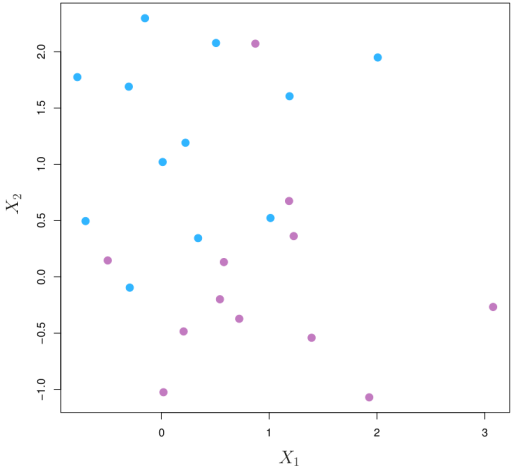
$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}) \geq M \;\; \forall \, i = 1, \ldots, n$$

- M is making sure every point is at least distance M from the hyperplane
- Because of the y_i multiplied, this also requires that the point is on the correct side of the hyperplane
- Last eq forces points to be on the correct side of the hyperplane if $M > 0$
- Making M as big as possible is the maximal margin hyperplane

Section 2

Issues with Maximal Margin Classifier

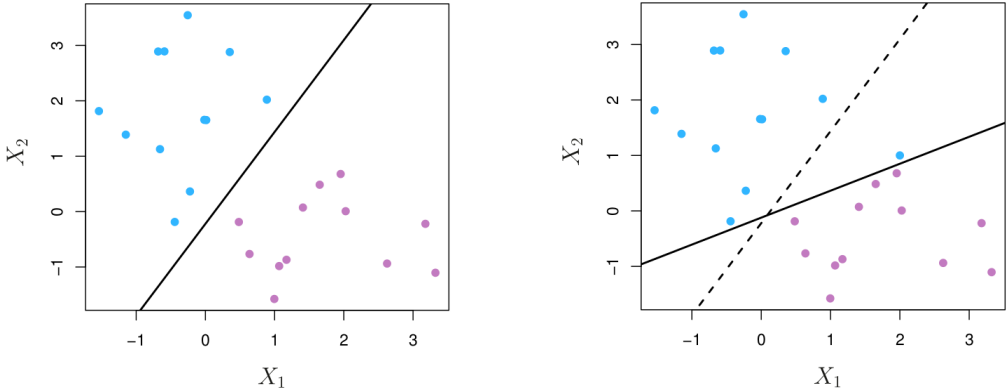
But what if....



- *...there is no separating hyperplane*
- *Means that there is no solution for* $M > 0$

Sensitivity to new points

Even if there is a spearating hyperplane might not be good



- Add a blue observation on the right
- Dramatic shift in maximal margin hyperplane

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